

Evaluation of Contextual Understanding in Large Language Models

Chamath Gunapala¹ §, Subavarshana Arumugam¹ §, Kithuni Wickramasinghe¹ §, Mamta Nallaretnam¹ §,
Pragatheeswaran Vipulanandan², Uthayasanker Thayasivam¹, Kamal Premaratne²

¹Department of Computer Science and Engineering, University of Moratuwa, Sri Lanka

²Department of Electrical and Computer Engineering, University of Miami, Coral Gables, Florida, USA
{chamathg.21, subavarshanaa.21, kithuni.21, nallaretnam}@cse.mrt.ac.lk,

Abstract—Large Language Models (LLMs) demonstrate impressive performance across diverse NLP tasks, yet their ability to exhibit genuine contextual understanding remains uncertain. Contextual understanding in LLMs refers to the ability to correctly extract relevant information from a given context, integrate it into a coherent internal representation, and reason over it to produce factually consistent and contextually grounded responses. However, traditional evaluation metrics such as perplexity, BLEU, or surface-level accuracy fail to reveal how well LLMs extract, integrate, and reason over contextual information. This gap is particularly critical in question answering (QA), where models must align responses with contextually grounded knowledge rather than relying on memorized associations. This paper proposes a novel knowledge graph (KG)-based evaluation framework to systematically assess the contextual understanding of large language models (LLMs) in question answering (QA). To this end, a hybrid similarity measure, S3KG, is introduced, which integrates structural and semantic similarity to produce a continuous evaluation score. In addition, a diagnostic analysis framework is developed to identify and categorize reasoning errors, enabling fine-grained insights into model behavior. Together, these components establish a reproducible evaluation pipeline for measuring correctness, faithfulness, and interpretability in LLM-generated responses, providing a transparent and scalable methodology for analyzing and improving contextual comprehension in LLMs.

Index Terms—large language models, knowledge graphs, contextual understanding, graph similarity

I. INTRODUCTION

Large Language Models (LLMs), such as Bidirectional Encoder Representations from Transformers (BERT), the Generative Pre-trained Transformer (GPT) models, and their advanced successors, have fundamentally transformed natural language processing (NLP). These models exhibit extraordinary proficiency in generating coherent, human-like text, answering complex questions, and executing a broad spectrum of tasks[?]. However, a significant challenge remains: determining whether these models genuinely understand the content they process or merely rely on sophisticated statistical correlations to produce plausible outputs.

Traditional evaluation metrics—such as perplexity, Bilingual Evaluation Understudy (BLEU) [1], and token-level matching on standardized benchmarks primarily measure surface level performance and fail to capture the depth of semantic comprehension or contextual understanding. This limitation

raises critical concerns about the reliability of LLMs in high-stakes scenarios such as educational tools, medical question answering systems, or legal analysis.

II. RELATED WORK

A. Evaluation of Contextual Understanding in LLMs

Evaluating how effectively LLMs understand and utilize contextual information remains a key challenge in QA systems, as fluent and plausible responses do not necessarily reflect faithful use of the provided context [2]. Zhu et al. [3] evaluate LLMs across four core tasks: coreference resolution, discourse relation classification, dialogue state tracking, and query rewriting, showing that while LLMs capture general context patterns, they fail to reveal which parts of the context are misunderstood and have not been evaluated in the QA domain. Complementing this, Yan et al. [4] probe LLM reasoning by manipulating in-context examples, including logical modifications (e.g., swapping “AND”/“OR”), finding that models do not consistently obey formal reasoning rules nor exhibit clearly identifiable error patterns. Together, these works highlight that current LLMs demonstrate strong surface-level comprehension but still struggle with fine-grained context interpretation and logical reasoning; evaluating these limitations in long-form LLM answers is particularly challenging, making knowledge graphs, which requires accurate extraction of relational triplets from text, a promising evaluation approach.

B. Open-domain Relational Triplet Extraction

Early approaches to Knowledge Graph construction treated the problem as a two-stage supervised pipeline, (1) identifying named entities and (2) classifying the relations between them, over manually defined schemas [5, 6], which prevented generalization beyond fixed ontologies. The introduction of pretrained language models shifted the field toward joint extraction architectures [7], giving rise to both discriminative [8] and generative [9] formulations that handled entities and relations in a single pass, though schema dependency remained a fundamental constraint. The move toward open-domain extraction gained traction with LLMs [10], giving rise to construction frameworks such as GraphRAG [11] and SAC-KG [12], as well as structured prompting pipelines [13] and iterative refinement approaches [14]; however, these were

§These authors contributed equally to this work

designed for single-document settings and produce inconsistent labels across multiple texts, making them poorly suited for cross-graph comparison. Recent hallucination evaluation frameworks such as [15] and [16] address this through few-shot prompting with instruction tuning to encourage label consistency across constructed graphs.

C. Graph Similarity Methods

Knowledge graph similarity has been approached through both embedding-based and kernel-based methods, each offering complementary strengths.

Bordes et al. [17] introduced TransE, a foundational embedding model that represents relations as translations in a continuous vector space, such that for a valid triple (h, r, t) the relation embedding satisfies $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$. While computationally efficient and effective for one-to-one relations, TransE cannot model symmetric or compositional relational patterns. Sun et al. [18] address this by representing each relation as a rotation in complex vector space under the constraint $|\mathbf{r}_i| = 1$, enabling the model to capture symmetry, antisymmetry, inversion, and composition within a unified formulation. Both approaches learn entity and relation embeddings over a fixed vocabulary, making them well-suited for link prediction on static KGs but ill-suited for cross-graph similarity tasks where entity sets are disjoint and no shared training corpus exists.

Graph kernels offer a training-free alternative for structural comparison. The Weisfeiler-Lehman (WL) graph kernel [19] iteratively aggregates node neighbourhood labels across h hops, producing a label histogram that can be compared via inner product to yield a structural similarity score. Togninalli et al. [20] extend this framework by replacing histogram comparison with the Wasserstein distance between WL-refined node embedding distributions, yielding the Wasserstein Weisfeiler-Lehman (WWL) kernel, which better accommodates continuous node attributes and reduces sensitivity to structural noise. A key limitation shared by both approaches is that labels are treated as opaque symbols: semantically equivalent but lexically distinct labels are considered entirely disjoint, reducing kernel sensitivity when applied to independently constructed knowledge graphs whose surface forms may differ.

Haskins and Adams [16] address this limitation through a neuro-symbolic framework that combines WL graph kernel computation with SBERT-based [21] semantic clustering of entity and relation labels prior to kernel evaluation. By grouping labels whose cosine similarity exceeds a fixed threshold, KEA enables structural comparison across KGs with heterogeneous surface forms. Sentence-BERT [21], trained via a siamese network with a contrastive objective, produces fixed-size sentence embeddings that support scalable pairwise semantic matching through cosine distance. While effective for hallucination detection, KEA’s clustering-based alignment is lossy: entities with similar but non-identical semantics may be merged, discarding fine-grained relational distinctions. The proposed S3KG method addresses this through type-separated, one-to-one label alignment, preserving entity-relation distinc-

tions while retaining semantic flexibility across independently constructed graphs.

III. METHODOLOGY

Our framework evaluates LLM contextual understanding using a knowledge graph (KG)-based pipeline, as shown in Fig.1. Given a question and its supporting context, an LLM generates a response, from which KGs are constructed alongside those derived from the ground truth and context. These graphs are then compared to measure similarity, and low-scoring cases are further analyzed using a triplet-level diagnostic module to identify reasoning errors.

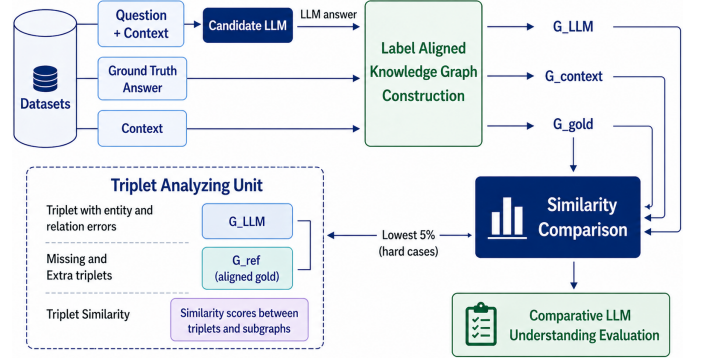


Fig. 1. Methodology pipeline for comparative LLM evaluation

A. LLM Answer Collection

A set of instruction-tuned open-source language models of comparable parameter scale are evaluated to ensure fair comparison without the influence of model size. Each model is prompted with a question and its associated supporting context, and the model-generated response is recorded alongside the reference answer for evaluation. To establish a baseline, responses are first generated deterministically by setting temperature to zero, yielding outputs that closely adhere to the provided context. Additional responses are then generated at progressively higher temperature settings to examine how increasing generation diversity influences the model’s ability to retain and utilize contextual information.

B. Knowledge Graph Construction

Three sources of information for each question, are transformed into structured knowledge graphs: the gold standard answers, the model-generated responses, and the supporting contexts. To ensure that the resulting graphs are comparable, we adopt the single few-shot prompting strategy with instruction tuning used in [15] and [16], applying a shared extraction prompt uniformly across all three sources. This ensures that entities and relations are extracted under the same schema, encouraging a consistent label space across all three graphs from the outset.

Following extraction, an additional NLP normalization step is applied uniformly to every entity and relation label across all three graphs. This includes lowercasing, lemmatization, and

whitespace normalization, ensuring that any residual surface-level variation introduced during extraction does not persist into the final graph representations. Together, the consistency-aware prompting and post-extraction normalization ensure that the three graphs are structurally compatible and ready for meaningful comparison.

C. S3KG: Semantic Structural Similarity for KGs

S3KG computes the similarity between two knowledge graphs G_1 and G_2 , each represented as a set of triples $\mathcal{T}_1 = \{(h_i, r_i, t_i)\}$ and $\mathcal{T}_2 = \{(h_j, r_j, t_j)\}$, where h , r , and t denote the head entity, relation, and tail entity, respectively. Rather than comparing graphs by a single criterion, S3KG operates at two levels: at the *triple level*, individual facts are matched to assess fine-grained correspondence between specific relational assertions; at the *graph level*, the overall topology is evaluated to capture how the entire relational structure aligns. At each level, two types of similarity are considered: *structural similarity*, which measures how closely the graph topologies match, and *semantic similarity*, which measures meaning level correspondence between entities and relations. This four way decomposition is realised through the following pipeline.

Triple-Level Matching. Each triple is serialised as a natural language string. SBERT embeddings [21] are computed for all triples in both sets using `paraphrase-MPNet-base-v2`. For each triple in $\mathcal{T}_1$, the most semantically similar triple in $\mathcal{T}_2$ is selected by cosine similarity, producing a filtered set $\hat{\mathcal{T}}_2 \subseteq \mathcal{T}_2$ that anchors the comparison to semantically relevant content.

Soft Label Alignment. Standard WL kernels assign zero credit to semantically equivalent labels with different surface forms. S3KG resolves this by aligning entity and relation labels independently before kernel computation. For each label ℓ in G_1 , if its maximum cosine similarity to any label in G_2 exceeds threshold $\tau = 0.65$, it is mapped to the best-matching label $\ell_{G_2}^*$:

$$\hat{\ell} = \begin{cases} \ell_{G_2}^* & \text{if } \max_j \cos(\phi(\ell), \phi(\ell_j)) \geq \tau \\ \ell & \text{otherwise} \end{cases} \quad (1)$$

where $\phi(\cdot)$ denotes the SBERT embedding. Entity and relation labels are aligned separately to prevent cross-type collisions during kernel evaluation.

WL Kernel Structural Similarity. Let $\tilde{G}_1$ and $\tilde{G}_2$ denote the soft-label-aligned versions of G_1 and G_2 produced by the preceding alignment step, and let $K(\cdot, \cdot)$ denote the Weisfeiler-Lehman (WL) graph kernel [19]. With 5 iterations, each node label is updated by aggregating its own label with those of its neighbours and their connecting relation labels, progressively encoding multi-hop structural patterns. The normalised structural similarity is:

$$S_{WL}(\tilde{G}_1, \tilde{G}_2) = \frac{K(\tilde{G}_1, \tilde{G}_2)}{\sqrt{K(\tilde{G}_1, \tilde{G}_1) \cdot K(\tilde{G}_2, \tilde{G}_2)}} \in [0, 1] \quad (2)$$

SBERT Mean-Pool Semantic Similarity. Each triple $(h, r, t) \in \mathcal{T}$ is encoded by SBERT into an embedding $\phi(h, r, t)$; the graph-level representation is the mean $\bar{e}_{\mathcal{T}} = \frac{1}{|\mathcal{T}|} \sum_{(h,r,t) \in \mathcal{T}} \phi(h, r, t)$. Semantic similarity is the cosine between the pooled representations of $\mathcal{T}_1$ and $\hat{\mathcal{T}}_2$, clipped to $[0, 1]$:

$$S_{SBERT}(\mathcal{T}_1, \hat{\mathcal{T}}_2) = \max\left(0, \frac{\bar{e}_{\mathcal{T}_1} \cdot \bar{e}_{\hat{\mathcal{T}}_2}}{\|\bar{e}_{\mathcal{T}_1}\| \|\bar{e}_{\hat{\mathcal{T}}_2}\|}\right) \quad (3)$$

This captures sentence-level meaning that discrete WL label refinement cannot express.

Final Score. The structural and semantic scores are combined via a mixing coefficient $\alpha \in [0, 1]$:

$$S_{S3KG} = (1 - \alpha) S_{WL} + \alpha S_{SBERT} \quad (4)$$

At $\alpha = 0.5$, the score equally weights structural fidelity from WL neighbourhood aggregation over aligned labels and semantic coherence from SBERT mean-pooling, encoding both local relational patterns and global meaning within a single continuous score.

D. Triplets Analyzing Unit

For the lowest-scoring 5% of QA pairs, we apply a triplet analysis unit to identify where and how the LLM generated KG diverges from the reference. Each triplet is converted into a natural language sentence and encoded using a Sentence Transformer model. Cosine similarity is computed between ground-truth and LLM triplet embeddings; pairs exceeding threshold 0.65–0.80 are treated as aligned triplets.

After removing aligned triplets, residual pairs are categorized into interpretable error classes based on component wise cosine similarities for head, relation, and tail:

- 1) Relation wrong: entities match but relation diverges.
- 2) Entity wrong: relation aligns but entity pair is inconsistent.
- 3) Extra Triplets: extra (hallucinated) triplets in LLM.
- 4) Missing Triplets: triplets that are not captured by the LLM.

IV. EXPERIMENTAL RESULTS

A. Datasets

Two context-rich QA datasets containing long-form answers are used for evaluation. PubMedQA [22] contains 273,518 biomedical question-answer pairs drawn from research articles, with answers typically exceeding 100 words, providing a testbed for domain-specific detailed response evaluation. MESAQA [23] comprises approximately 6,100 question-answer pairs from consumer healthcare documents, featuring abstractive answers averaging 70 words that require multi-span evidence integration. Together, the two datasets provide evaluation coverage across academic biomedical reasoning and practical healthcare knowledge synthesis.

B. LLM Answer Collection

Three instruction-tuned open-source language models with 7-billion parameters are evaluated: Llama-2-7b-chat-hf, Gemma-7b-it, and Mistral-7B-Instruct-v0.2. These models are selected because they operate at a similar scale, allowing for fair comparison without the influence of model size. Each model is prompted with a question and its associated supporting context, and the model-generated response is recorded alongside the reference answer for evaluation. To establish a baseline, responses are first generated deterministically at temperature 0, yielding outputs that closely adhere to the provided context. Additional responses are then generated at higher temperature settings (0.3, 0.7, and 1.0) to examine how increasing generation diversity influences the model’s ability to retain and utilize contextual information.

C. Benchmarking Dataset Collection

To assess generalisation across diverse text types, we evaluate on ten datasets spanning three structural categories. Each dataset is cast as a binary classification task: given a text pair (s_1, s_2) , predict whether the pair is semantically equivalent (label = 1) or not (label = 0). All datasets are balanced at 200 positive and 200 negative pairs ($N = 400$). Performance is measured by maximum F1 score, obtained via threshold sweep, and AUROC.

1) *Short-Text with Human-Annotated*: Three datasets contain sentence pairs averaging 10–22 words with crowd-sourced or expert equivalence labels: **MRPC** [24], consisting of news sentence pairs with paraphrase labels; **PAWS-Wiki** [25], adversarially constructed paraphrase pairs from Wikipedia where lexical overlap is deliberately an unreliable signal; and **STS12** [26], sentence similarity pairs drawn from multiple NLP tasks.

2) *KG-Perturbed Paragraphs*: Six datasets contain paragraphs averaging 69–126 words, constructed by perturbing entity relationships in KG-derived paragraph representations. The five evaluated datasets, **SK-Codex 400**, **SK-Combined**, **SK-FindKG**, **SK-GloBI**, and **SK-Oregano**, differ in their underlying knowledge graph source, covering general encyclopaedic (Codex [27]), financial/economic (FindKG), biological interaction (GloBI [28]), and food ontology (Oregano) domains.

3) *NLP-Perturbed Paragraphs (Anti-Circularity Control)*: The **Wikipedia Entity-Swap** dataset is constructed by replacing named entities in Wikipedia passages entirely independently of any KG pipeline. It serves as a critical anti-circularity test: if our method’s gains were an artefact of circular evaluation i.e., performing well only because the test data was generated by the same KG pipeline that produces the embeddings performance on this independently constructed dataset should collapse.

D. Methods Evaluated

S3KG is evaluated with a mixing coefficient α which controls the blend between KG structural signal and sentence-transformer signal defined in Eq. 4, such that $\alpha = 0.0$ recovers

TABLE I
PERFORMANCE ON SHORT-TEXT DATASETS (F1 / AUROC).

Method	MRPC	PAWS-Wiki	STS12
S3KG	0.692 / 0.673	0.766 / 0.795	0.786 / 0.834
ROUGE-1	0.745 / 0.784	0.678 / 0.490	0.725 / 0.754
ROUGE-2	0.720 / 0.721	0.715 / 0.721	0.681 / 0.656
ROUGE-L	0.729 / 0.760	0.735 / 0.807	0.703 / 0.710
BLEU	0.687 / 0.677	0.716 / 0.747	0.671 / 0.644
BERTScore	0.758 / 0.816	0.691 / 0.702	0.682 / 0.636
MiniLM	0.723 / 0.748	0.687 / 0.638	0.833 / 0.894
sentence-T5-base	0.766 / 0.816	0.674 / 0.668	0.853 / 0.928

a pure KG structural embedding and $\alpha = 1.0$ recovers a pure dense sentence-transformer representation. Pure KG structural embeddings capture relational and ontological structure between concepts but lack lexical and semantic expressiveness, whereas sentence transformers excel at surface-level semantic similarity yet remain blind to the underlying graph topology. S3KG bridges this gap by interpolating between both signals, enabling richer matching that is sensitive to both conceptual structure and natural language meaning. For each dataset, we report the best-performing variant selected by maximum F1 across the sweep $\alpha \in \{0.0, 0.1, \dots, 1.0\}$; the full per-dataset α sweep results are provided in the Appendix. We compare against seven standard baselines: ROUGE-1, ROUGE-2, ROUGE-L [29], BLEU [1], BERTScore [30], MiniLM [31], and sentence-T5-base [32].

E. Short-Text Dataset Results

Table I reports performance on the three short-text datasets. Best values per column are shown in **bold**.

On MRPC, S3KG peaks at (F1 = 0.692) falls below sentence-T5-base (F1 = 0.766), a gap that is consistent with the 10-word average sentence length yielding sparse KG triples. On PAWS-Wiki, the structural advantage of S3KG is more apparent: it achieves F1 = 0.766 while ROUGE-1 collapses to an AUC of 0.490 (near-random), confirming the well-known failure of n -gram overlap on adversarially constructed paraphrases. On STS12, S3KG achieves F1 = 0.786 (AUC = 0.834); the modest improvement over pure dense embedding suggests that even a small KG contribution is beneficial, though all variants remain below sentence-T5-base (F1 = 0.853).

F. Worked Example: Contextual Understanding via KG Structure

To illustrate how S3KG captures contextual meaning where surface-overlap metrics fail, we trace two pairs from the PAWS-Wiki dataset through the full pipeline. Each pair has near-identical surface form, making them adversarial stress tests for n -gram methods.

In Pair A, both sentences express the same proposition with only word-order variation. The KG pipeline extracts identical triples from both, yielding $\text{sim}_{\text{KG}} = 1.0$, and S3KG correctly predicts *Similar*. In Pair B, the subject and object of the winning relation are swapped between s_1 and s_2 : one

TABLE II

S3KG PIPELINE SCORES FOR TWO PAWS-WIKI SENTENCE PAIRS. IN THE POSITIVE EXAMPLE, WORD-ORDER VARIATION YIELDS IDENTICAL KG TRIPLETS AND $\text{sim}_\alpha = 1.0$, CORRECTLY PREDICTING *Similar*. IN THE NEGATIVE EXAMPLE, REVERSED SUBJECT–OBJECT ROLES IN THE TRIPLETS EXPOSE SEMANTIC OPPOSITION DESPITE IDENTICAL SURFACE TOKENS — WHERE ROUGE-1 FAILS BY PREDICTING *Similar*, S3KG CORRECTLY PREDICTS *Not Similar*.

	Positive Example	Negative Example
Text s_1	<i>His father returned as a finished violinist of the Russian School to Bombay.</i>	<i>Renzo Furlan won 6–3, 6–4 against Thomas Johansson in the finals.</i>
Text s_2	<i>His father returned to Bombay as a finished violinist of the Russian school.</i>	<i>Thomas Johansson won 6–3, 6–4 against Renzo Furlan in the finals.</i>
Triplet (s_1)	(father, returned_to, Bombay)	(Renzo Furlan, won_against, Thomas Johansson)
Triplet (s_2)	(father, returned_to, Bombay)	(Thomas Johansson, won_against, Renzo Furlan)
sim_{TP} ($\alpha=0$)	1.0000	0.8310
sim_{ST} ($\alpha=1$)	1.0000	0.1361
sim_α ($\alpha=0.5$)	1.0000	0.4835
Threshold τ	0.92	0.92
S3KG pred.	Similar ✓	Not Similar ✓
ROUGE-1 score	1.0000	1.0000
ROUGE-1 pred.	Similar ✓	Similar ×
True label	Positive (1)	Negative (0)

sentence states Furlan defeated Johansson, the other states the reverse. Because every token appears in both sentences, ROUGE-1 scores 1.0 for both pairs and incorrectly predicts *Similar* for Pair B. S3KG, by contrast, extracts the directed triple (**subject**, won_against, **object**) and detects the reversal via structural graph divergence, correctly predicting *Not Similar*. This illustrates why KG-structural embeddings capture relational semantics that n -gram overlap metrics cannot.

G. KG-Perturbed Paragraph Dataset Results

Table III reports performance across the five KG-perturbed paragraph datasets.

On SK-Codex 400, S3KG achieves the strongest result across all methods ($F1 = 0.932$, $AUC = 0.973$), exceeding the best baseline, MiniLM ($F1 = 0.875$), by +5.7 F1 points, a substantial margin that highlights the advantage of structural KG embeddings when entity relationships are well-defined and consistently extractable.

On SK-Combined, S3KG leads ($F1 = 0.834$), outperforming sentence-T5-base ($F1 = 0.770$) by +6.4 points. The consistent improvement across mixed entity domains suggests that S3KG generalises beyond single-domain KGs, a property that purely lexical or dense sentence-level methods do not reliably exhibit.

In SK-FindKG, sentence-T5-base ($F1 = 0.848$) leads all methods, with S3KG achieving $F1 = 0.767$. We attribute this to KG extraction noise introduced by the financial and economic vocabulary characteristic of this source, where noisy triplets degrade the structural signal that S3KG relies on, underscoring that KG extraction quality is a meaningful bottleneck for structure-based approaches.

On SK-GloBI, S3KG achieves $F1 = 0.892$ ($AUC = 0.935$), exceeding BERTScore (0.816 / 0.871) by +7.6 F1 points. The biological interaction triples produced by GloBI are highly distinctive, making this the dataset most naturally aligned

with the structural embedding approach and confirming that relational specificity directly amplifies S3KG’s advantage over dense baselines.

On SK-Oregano, S3KG achieves $F1 = 0.812$ ($AUC = 0.892$), marginally outperforming sentence-T5-base ($F1 = 0.800$) by +1.2 points. While the margin is modest, it remains competitive, consistent with a food ontology providing moderate rather than high relational distinctiveness, suggesting performance scales with the structural richness of the underlying KG source.

H. Wikipedia Entity-Swap: Anti-Circularity Control

Table IV reports performance on the Wikipedia Entity-Swap dataset. ROUGE-1 achieves a perfect score due to a known dataset artefact: entity-swapped pairs differ only in the swapped entity tokens while sharing nearly identical surrounding surface-form tokens, making unigram overlap trivially near-perfect. ROUGE-1 is therefore excluded from the meaningful comparison.

S3KG ($F1 = 0.872$, $AUC = 0.890$) achieves the best meaningful result among all methods, outperforming all baselines except the artefact-driven ROUGE-1. The performance is consistently higher than MiniLM ($F1 = 0.821$) and sentence-T5-base ($F1 = 0.762$), confirming that KG-based methods generalise to independently constructed evaluation data and refuting circularity as an explanation for the observed gains.

I. Cross-Dataset Analysis

Table V summarises the best S3KG variant and comparative verdict for each dataset.

Three patterns emerge from the cross-dataset view.

Text richness drives KG performance. Performance of KG-based variants scales with text length and relational density. On short texts (10–22 words) the structural KG signal is sparse and our methods are competitive but not dominant.

TABLE III
PERFORMANCE ON KG-PERTURBED PARAGRAPH DATASETS (F1 / AUROC).

Method	C400	Comb.	Find	GloBI	Oreg.
S3KG	0.932/0.973	0.834/0.829	0.767/0.796	0.892/0.935	0.812/0.892
ROUGE-1	0.835/0.917	0.732/0.728	0.745/0.745	0.784/0.833	0.745/0.782
ROUGE-2	0.822/0.894	0.707/0.711	0.717/0.706	0.776/0.833	0.752/0.791
ROUGE-L	0.792/0.855	0.722/0.717	0.719/0.721	0.763/0.800	0.792/0.835
BLEU	0.806/0.884	0.715/0.708	0.711/0.710	0.775/0.819	0.745/0.794
BERTScore	0.823/0.916	0.757/0.792	0.739/0.761	0.816/0.871	0.743/0.798
MiniLM	0.875/0.943	0.770/0.789	0.802/0.844	0.780/0.817	0.773/0.814
sent-T5	0.876/0.944	0.770/0.827	0.848/0.902	0.728/0.760	0.797/0.871

C400: SK-Codex 400; Comb.: SK-Combined; Find: SK-FindKG; Oreg.: SK-Oregano.

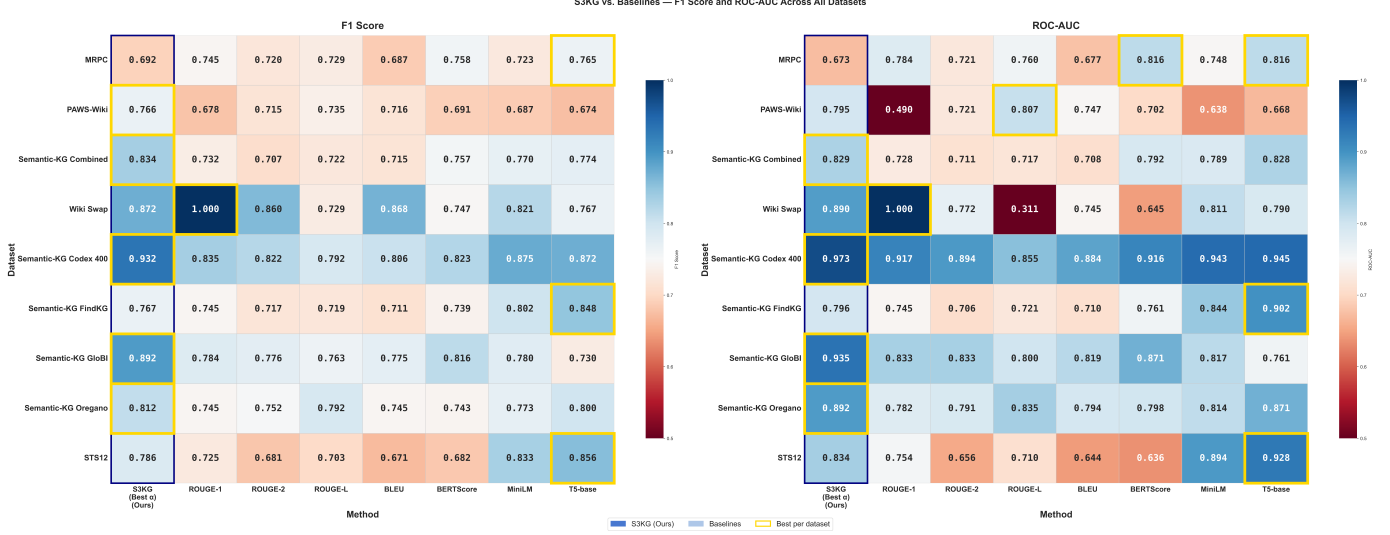


Fig. 2. F1 Score (left) and AUROC (right) of S3KG and baseline methods across all benchmark datasets. Gold borders indicate the best-performing method per dataset. S3KG (navy border, leftmost column) represents the best-performing α variant selected per dataset from the full α sweep (see Appendix).

TABLE IV
Results on Wikipedia Entity-Swap ($N = 400$). The proposed S3KG achieves the highest F1 and AUC, outperforming all lexical and neural baselines.

Method	F1	AUC	Prec.	Rec.
S3KG	0.872	0.890	0.780	0.990
ROUGE-2	0.860	0.772	0.773	0.970
ROUGE-L	0.729	0.311	0.573	1.000
BLEU	0.868	0.745	0.776	0.985
BERTScore	0.747	0.645	0.615	0.950
MiniLM	0.821	0.811	0.729	0.940
sentence-T5-base	0.762	0.806	0.621	0.985

TABLE V
Best S3KG variant per dataset, selected by maximum F1 via grid search over $\alpha \in \{0.0, 0.1, \dots, 1.0\}$. KG-rich datasets gain up to +7.6 F1 over the best baseline; sparse or noisy KGs yield reduced margins.

Dataset	F1 / AU-ROC	Verdict
MRPC ($\alpha = 0.3$)	0.692 / 0.673	Behind (sparse KG)
PAWS-Wiki ($\alpha = 0.5$)	0.766 / 0.795	Strong (+3.1 F1)
STS12 ($\alpha = 0.1$)	0.786 / 0.834	Behind (short text)
SK-Codex 400 ($\alpha = 0.5$)	0.932 / 0.973	Strong (+5.7 F1)
SK-Combined ($\alpha = 0.5$)	0.834 / 0.829	Strong (+6.0 F1)
SK-FindKG ($\alpha = 0.0$)	0.767 / 0.796	Behind (domain noise)
SK-GloBI ($\alpha = 0.6$)	0.892 / 0.935	Strong (+7.6 F1)
SK-Oregano ($\alpha = 0.4$)	0.812 / 0.892	Comparable (+1.2)
Wiki Swap ($\alpha = 0.1$)	0.872 / 0.890	Best meaningful score

On paragraph-level datasets (69–126 words), S3KG variants achieve top-1 or top-2 performance on 7 of 8 benchmarks, with F1 improvements of 3–12 points over the strongest baseline.

The optimal α is dataset-dependent. The best-performing α varies across datasets, lower values benefit datasets where perturbations are primarily structural (Wiki Swap, SK-FindKG), while higher values are best where KG and dense signals are complementary (SK-Codex 400, SK-GloBI, SK-Combined.)

The complete α sweep results are reported in the Appendix. *Baseline behaviour is dataset-dependent.* Token-overlap baselines (ROUGE, BLEU) are strong on KG-perturbed datasets where perturbations alter surface form, but unreliable on adversarial datasets (PAWS-Wiki: ROUGE-1 AUC = 0.490; Wiki Swap: ROUGE-L AUC = 0.311). BERTScore is more stable but consistently underperforms S3KG on KG-

TABLE VI
TRIPLET ANALYZING UNIT PERFORMANCE

Metric	KEA Baseline	Ours
Micro Precision	0.853	0.847
Micro Recall	0.738	0.949
Macro Precision	0.953	0.803
Macro Recall	0.641	0.946
Micro F1	0.836	0.895
Macro F1	0.622	0.782

perturbed data. Dense sentence embedding models (MiniLM, sentence-T5-base) are the strongest baselines overall but require full fine-tuned transformer inference, whereas S3KG’s KG component is comparatively lightweight at inference time.

J. Diagnosing Context and Answer Deviations

Dataset and Annotation: We evaluate the triplet analyzing unit on a manually annotated subset of the **MessaQA** dataset, which contains general health-related questions with long form answers. For each QA instance, two knowledge graphs are constructed: (i) a ground-truth KG extracted from the reference answer, and (ii) an LLM generated (Llama and Gamma was used) KG extracted from the model output. The task of the triplet analysis unit is to identify semantically aligned triplet pairs between these two graphs. To obtain reliable evaluation labels, we created a gold set of aligned triplet pairs (GT $\leftrightarrow$ LLM). Alignment was independently annotated by three medical students following a fixed guideline defining semantic equivalence at the triplet level. Disagreements were resolved through adjudication.

Annotation Quality: Inter annotator agreement was measured over candidate aligned triplet pairs. The results indicate strong consistency, with percent agreement of 0.9433, pairwise F1 scores of 0.9708, 0.9825, and 0.9882, and pairwise Jaccard scores of 0.9433, 0.9657, and 0.9766, confirming the reliability of the annotations.

Evaluation Protocol: Performance is evaluated against the annotated alignments using precision, recall, and F1 score, reported using both micro averaged metrics (aggregated over all triplets) and macro averaged metrics (computed per QA instance and averaged), capturing both overall performance and consistency across samples.

Results: As shown in Table VI, the KEA baseline achieves higher precision due to its conservative component wise matching, but exhibits low recall, missing many valid alignments. In contrast, our sentence-level alignment approach significantly improves recall (+34.8%), resulting in higher Micro and Macro F1 scores. This improvement arises from robustness to lexical variation in relations. For example, semantically equivalent relations such as “*treats*” and “*used for*” are often not aligned by KEA, whereas our method captures such equivalence through sentence level semantic similarity. The lower precision reflects the expected trade-off when moving from strict lexical matching to semantic matching, while the overall F1 improvement indicates a better balance between sensitivity and specificity. Additionally, our method

enables residual error analysis by categorizing mismatches into relation wrong and entity wrong types.

K. LLM Comparison on QA Datasets

Having validated S3KG as a reliable graph similarity measure, we apply it as an evaluation instrument to address the following research question: *given a question and its supporting context, to what extent does an LLM capture the relational knowledge of the reference answer, and how faithfully does its response reflect the provided context?* Traditional metrics such as BLEU or exact match are insufficient for this purpose, as they assess surface-level token overlap rather than the relational knowledge structure of a response.

To operationalise this, three knowledge graphs are constructed per QA instance i : $KG_{\text{LLM}}^{(i)}$ from the model-generated response, $KG_{\text{gold}}^{(i)}$ from the reference answer, and $KG_{\text{ctx}}^{(i)}$ from the supporting context. S3KG is then applied to compute two independent similarity scores. **Gold Similarity** is defined as $\text{GoldSim}(i) = S_{\text{S3KG}}(KG_{\text{LLM}}^{(i)}, KG_{\text{gold}}^{(i)})$, measuring factual accuracy: a higher value indicates that the model’s response aligns more closely with the relational structure of the ground truth answer. **Context Similarity** is defined as $\text{CtxSim}(i) = S_{\text{S3KG}}(KG_{\text{LLM}}^{(i)}, KG_{\text{ctx}}^{(i)})$, measuring contextual faithfulness: a higher value indicates that the model draws more effectively on the provided context rather than relying on parametric knowledge. These two dimensions are complementary and can diverge - a model may closely reflect the context while deviating from the reference facts, or vice versa, thereby exposing capability trade-offs that a single aggregate metric would conceal.

To produce a single ranked score, we define the **Contextual Understanding Score (CUS)**. For each QA instance $i \in \{1, \dots, N\}$, where N is the total number of evaluation samples, the per-sample CUS is the harmonic mean of GoldSim and CtxSim:

$$\text{CUS}(i) = \frac{2 \cdot \text{GoldSim}(i) \cdot \text{CtxSim}(i)}{\text{GoldSim}(i) + \text{CtxSim}(i)} \quad (5)$$

The harmonic mean penalises imbalanced profiles: a model that is highly contextually faithful but factually imprecise or vice versa, is ranked below a model that achieves moderate scores on both dimensions. The dataset-level score is then the mean of $\text{CUS}(i)$ over all N samples:

$$\text{CUS} = \frac{1}{N} \sum_{i=1}^N \text{CUS}(i) \quad (6)$$

To assess these dimensions, four 7B-parameter models are evaluated on both PubMedQA and MesaQA using S3KG with $\alpha = 0.5$, with mean GoldSim, CtxSim, and CUS (Eq. 6) reported in Table VII for all model dataset combinations ($N=400$ per configuration).

MesaQA Results. On MesaQA, Gemma-7B, Llama-2-7B, and Mistral-7B perform similarly with CUS scores between 0.675 and 0.678, while Falcon-7B scores noticeably lower

TABLE VII
LLM EVALUATION RESULTS (MEAN OVER $N=400$ SAMPLES).

Dataset	Model	GoldSim	CtxSim	CUS
MesaQA	Gemma-7B	0.6889	0.7031	0.6774
	Llama-2-7B	0.6570	0.7236	0.6752
	Mistral-7B	0.6491	0.7356	0.6780
	Falcon-7B	0.6036	0.6458	0.6063
PubMedQA	Gemma-7B	0.5235	0.6401	0.5587
	Llama-2-7B	0.5220	0.6567	0.5651
	Mistral-7B	0.5138	0.7331	0.5923
	Falcon-7B	0.4541	0.5560	0.4800

(CUS 0.6063). Looking at the two dimensions separately, Gemma-7B scores highest on GoldSim (0.6889), meaning its answers are closest to the reference, while Mistral-7B leads on CtxSim (0.7356), showing stronger use of the provided context. Across all models, CtxSim is consistently higher than GoldSim, suggesting that models draw well from the context but add phrasing or information not present in the gold answer. Mistral-7B achieves the best overall CUS (0.6780) by balancing both dimensions effectively. Falcon-7B shows the weakest performance across all three metrics on MesaQA, indicating that it struggles to produce knowledge graphs that align with either the reference answer or the contextual information.

PubMedQA Results. Performance drops noticeably on PubMedQA, with CUS scores ranging from 0.480 (Falcon-7B) to 0.592 (Mistral-7B). GoldSim spans 0.45–0.52 and CtxSim spans 0.56–0.73, compared to 0.60–0.69 and 0.65–0.74 on MesaQA respectively, reflecting the difficulty of matching precise biomedical reference answers. Mistral-7B again leads in CUS (0.5923), supported by a strong CtxSim (0.7331). Among the top three models, Gemma-7B scores highest in GoldSim (0.5235) but lowest in CtxSim (0.6401), resulting in a lower CUS (0.5587); this indicates that Gemma-7B produces content closer to the reference answer but relies less on the provided biomedical context. Falcon-7B shows the weakest results across all metrics on PubMedQA, suggesting that biomedical domain knowledge presents particular challenges for this model.

Cross-Dataset Observations. CtxSim exceeds GoldSim in all eight model–dataset combinations, showing that all models align more closely with context documents than with gold answers. This is likely because instruction-tuned models tend to elaborate on context rather than produce concise reference-style responses. The consistent gap of around 15 percentage points in CUS between MesaQA and PubMedQA across the three stronger models (Gemma-7B, Llama-2-7B, Mistral-7B) points to domain complexity as the main factor—biomedical vocabulary and reasoning make contextual understanding harder regardless of model size or architecture. Falcon-7B exhibits a comparatively larger performance gap and lower absolute scores across both datasets, suggesting it is less well-suited for knowledge-graph-level evaluation on both general and biomedical QA tasks.

Triplet analyzing on the lowest similar Kgs Across

the lowest-similarity 5% cases in both datasets, the three models show clear differences. Mistral is consistently the most reliable on the hard examples, preserving more reference facts and producing more aligned triplets than the other models. Llama generally falls in the middle, while Gemma shows the weakest performance, especially on the general health QA set where it often fails to recover many gold-aligned facts. A second consistent pattern is that, in these difficult cases, the generated knowledge graphs tend to align more closely with the context knowledge graph than with the reference knowledge graph. This suggests that many failures are not simply random errors, but cases where the model either drifts toward a different interpretation, omits key reference facts, or introduces unsupported additions. Finally, the PubMed setting is noticeably harder for all models: aligned-triplet recovery drops sharply, indicating that technical biomedical terminology, abbreviations, and entity variability dominate the failure modes in the hardest cases.

V. DISCUSSION

The proposed KG based framework provides a structured approach to evaluate LLM understanding, moving beyond surface level metrics to assess relational and structural fidelity. S3KG’s type separated, one to one label alignment addresses fundamental limitations of clustering based approaches while maintaining the WL kernel’s ability to capture multi-hop neighborhood similarity.

The consistent performance gap between PubMedQA and MesaQA scores across all three models reveals a measurable difference in LLM capability for domain specific versus general health knowledge. The two dimensional S3KG evaluation further distinguishes factual accuracy (gold similarity) from contextual faithfulness (context similarity), exposing model specific trade offs that aggregate metrics cannot capture. The triplet analysis unit additionally enables pinpointing specific reasoning failures, whether due to incorrect entity substitution, relation errors, or broader inconsistencies.

Key limitations include: (1) sensitivity of similarity scores to KG extraction quality; (2) computational cost for large scale evaluation. Future work will incorporate larger proprietary models (GPT-4), attention-based alignment for directional semantic equivalence (e.g., *daughter_of* vs. *mother_of*), and ambiguous attention distribution handling.

VI. CONCLUSION

We presented a novel KG based evaluation framework for assessing LLM contextual understanding in QA. By constructing canonicalized knowledge graphs from LLM outputs, gold answers, and supporting context, and comparing them using S3KG, we move beyond surface level accuracy toward verifiable, graph theoretic comprehension measurement. S3KG achieves the highest overall correlation with ground-truth similarity (Spearman 0.899, MAE 0.105, RMSE 0.144) across five diverse benchmarks, ranking first among nine competing methods. The Triplet Analyzing Unit provides interpretable, fine-grained diagnostics of reasoning failures. Evaluations

on PubMedQA and MesaQA demonstrate consistent model-specific strengths and weaknesses, establishing a reproducible pipeline that can be extended to other datasets and tasks to support trustworthy AI development.

REFERENCES

- [1] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, “Bleu: A method for automatic evaluation of machine translation,” in *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, Philadelphia, PA, USA, 2002, pp. 311–318.
- [2] P. Vipulanandan, K. Premaratne, and D. Sarkar, “Semantic uncertainty quantification of hallucinations in llms: A quantum tensor network based method,” *arXiv preprint arXiv:2601.20026*, 2026.
- [3] Y. Zhu, J. R. A. Moniz, S. Bhargava, J. Lu, D. Piraviprumal, S. Li, Y. Zhang, H. Yu, and B.-H. Tseng, “Can large language models understand context?” in *Findings of the Association for Computational Linguistics: EACL 2024*, mar 2024, pp. 2004–2018.
- [4] J. Yan, C. Wang, J. Huang, and W. Zhang, “Do large language models understand logic or just mimic context?” *arXiv preprint arXiv:2402.12091*, 2024.
- [5] D. Zelenko, C. Aone, and A. Richardella, “Kernel methods for relation extraction,” *Journal of Machine Learning Research*, vol. 3, pp. 1083–1106, 2003.
- [6] B. M. Sundheim, “Overview of results of the MUC-6 evaluation,” in *Proceedings of the Sixth Message Understanding Conference (MUC-6)*, Columbia, Maryland, November 1995.
- [7] Y.-M. Shang, H. Huang, and X. Mao, “OneRel: Joint entity and relation extraction with one module in one step,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 10, June 2022, pp. 11 285–11 293.
- [8] Y. Wang, B. Yu, Y. Zhang, T. Liu, H. Zhu, and L. Sun, “TPLinker: Single-stage joint extraction of entities and relations through token pair linking,” in *Proceedings of the 28th International Conference on Computational Linguistics*, December 2020, pp. 1572–1582.
- [9] P.-L. H. Cabot and R. Navigli, “REBEL: Relation extraction by end-to-end language generation,” in *Findings of the Association for Computational Linguistics: EMNLP 2021*, November 2021, pp. 2370–2381.
- [10] X. Wei, X. Cui, N. Cheng, X. Wang, X. Zhang, S. Huang, P. Xie, J. Xu, Y. Chen, M. Zhang, and Y. Jiang, “ChatIE: Zero-shot information extraction via chatting with ChatGPT,” *arXiv preprint arXiv:2302.10205*, 2023.
- [11] D. Edge, H. Trinh, N. Cheng, J. Bradley, A. Chao, A. Mody, S. Truitt, D. Metropolitansky, R. O. Ness, and J. Larson, “From local to global: A graph RAG approach to query-focused summarization,” *arXiv preprint arXiv:2404.16130*, 2024.
- [12] H. Chen, X. Shen, Q. Lv, J. Wang, X. Ni, and J. Ye, “SAC-KG: Exploiting large language models as skilled automatic constructors for domain knowledge graph,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, August 2024, pp. 4345–4360.
- [13] B. Zhang and H. Soh, “Extract, define, canonicalize: An llm-based framework for knowledge graph construction,” in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024, pp. 9820–9836.
- [14] X. Jing, G. Wu, X. Zhuo, L. Sun, and J. Wang, “Knowledge restoration-driven prompt optimization: Unlocking llm potential for open-domain relational triplet extraction,” *arXiv preprint arXiv:2601.15037*, Jan. 2026.
- [15] H. Sansford, N. Richardson, H. P. Maretic, and J. N. Saada, “Grapheval: A knowledge-graph based llm hallucination evaluation framework,” *arXiv preprint arXiv:2407.10793*, 2024.
- [16] R. Haskins and B. Adams, “Kea explain: Explanations of hallucinations using graph kernel analysis,” *arXiv preprint arXiv:2507.03847*, 2025.
- [17] A. Bordes, N. Usunier, A. García-Durán, J. Weston, and O. Yakhnenko, “Translating embeddings for modeling multi-relational data,” in *Advances in Neural Information Processing Systems*, vol. 26. Curran Associates, Inc., 2013, pp. 2787–2795.
- [18] Z. Sun, Z.-H. Deng, J.-Y. Nie, and J. Tang, “RotatE: Knowledge graph embedding by relational rotation in complex space,” in *Proceedings of the 7th International Conference on Learning Representations (ICLR)*, 2019.
- [19] N. Shervashidze, P. Schweitzer, E. J. van Leeuwen, K. Mehlhorn, and K. M. Borgwardt, “Weisfeiler-Lehman graph kernels,” *Journal of Machine Learning Research*, vol. 12, pp. 2539–2561, 2011.
- [20] M. Togninalli, E. Ghisu, F. Llinares-López, B. Rieck, and K. Borgwardt, “Wasserstein Weisfeiler-Lehman graph kernels,” in *Advances in Neural Information Processing Systems*, vol. 32. Curran Associates, Inc., 2019, pp. 6439–6449.
- [21] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*. Hong Kong, China: Association for Computational Linguistics, 2019, pp. 3982–3992.
- [22] Q. Jin, B. Dhingra, Z. Liu, W. Cohen, and X. Lu, “Pubmedqa: A dataset for biomedical research question answering,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 2567–2577.
- [23] J.-I. Wang, H.-H. Huang, and H.-H. Chen, “MESAQA: A dataset for multi-span contextual and evidence-grounded question answering,” in *Proceedings of the 31st International Conference on Computational Linguistics*. Abu Dhabi, UAE: Association for

Computational Linguistics, Jan. 2025, pp. 10 891–10 901. [Online]. Available: <https://aclanthology.org/2025.coling-main.724/>

- [24] W. B. Dolan and C. Brockett, “Automatically constructing a corpus of sentential paraphrases,” in *Proceedings of the Third International Workshop on Paraphrasing (IWP2005)*, 2005.
- [25] Y. Zhang, J. Baldridge, and L. He, “PAWS: Paraphrase adversaries from word scrambling,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2019, pp. 1702–1711.
- [26] E. Agirre *et al.*, “SemEval-2012 task 6: A pilot on semantic textual similarity,” in *Proceedings of the First Joint Conference on Lexical and Computational Semantics (*SEM)*, 2012.
- [27] T. Safavi and D. Koutra, “Codex: A comprehensive knowledge graph completion benchmark,” in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021.
- [28] J. H. Poelen, J. D. Simons, and C. J. Mungall, “GloBI: Global biotic interactions,” [Online]. Available: <https://www.globalbioticinteractions.org>, 2014, accessed: Jan. 15, 2025.
- [29] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out: Proceedings of the ACL-04 Workshop*, 2004, pp. 74–81.
- [30] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *International Conference on Learning Representations (ICLR)*, 2020.
- [31] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, “MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5776–5788.
- [32] J. Ni, G. H. Ábrego, N. Constant, J. Ma, K. B. Hall, M. Chang, and Y. Yang, “Sentence-t5: Scalable sentence encoders from pre-trained text-to-text models,” in *Findings of the Association for Computational Linguistics: ACL 2022*, 2022, pp. 272–279.

APPENDIX

This appendix reports the complete S3KG α sweep results across all ten evaluation datasets. Each table shows F1 and AUROC for $\alpha \in \{0.0, 0.1, \dots, 1.0\}$, where $\alpha = 0.0$ recovers a pure KG structural embedding and $\alpha = 1.0$ recovers a pure dense sentence-transformer representation, as defined in Equation 4. The best-performing α per dataset (by F1) is shown in **bold**.

TABLE VIII
S3KG α SWEEP ON MRPC (F1 / AUROC).

α	F1	AUROC
0.0	0.676	0.654
0.1	0.681	0.664
0.2	0.683	0.669
0.3	0.692	0.673
0.4	0.680	0.674
0.5	0.680	0.675
0.6	0.681	0.674
0.7	0.688	0.674
0.8	0.684	0.672
0.9	0.683	0.671
1.0	0.681	0.671

TABLE IX
S3KG α SWEEP ON PAWS-WIKI (F1 / AUROC).

α	F1	AUROC
0.0	0.694	0.739
0.1	0.745	0.781
0.2	0.760	0.790
0.3	0.764	0.793
0.4	0.764	0.795
0.5	0.766	0.795
0.6	0.764	0.795
0.7	0.764	0.795
0.8	0.764	0.795
0.9	0.764	0.795
1.0	0.764	0.731

TABLE X
S3KG α SWEEP ON STS12 (F1 / AUROC).

α	F1	AUROC
0.0	0.783	0.828
0.1	0.786	0.834
0.2	0.780	0.834
0.3	0.780	0.831
0.4	0.780	0.827
0.5	0.775	0.824
0.6	0.768	0.820
0.7	0.764	0.815
0.8	0.762	0.810
0.9	0.756	0.805
1.0	0.756	0.790

TABLE XI
S3KG α SWEEP ON SK-CODEX 400 (F1 / AUROC).

α	F1	AUROC
0.0	0.871	0.969
0.1	0.922	0.973
0.2	0.927	0.973
0.3	0.927	0.973
0.4	0.929	0.973
0.5	0.932	0.973
0.6	0.932	0.973
0.7	0.932	0.973
0.8	0.932	0.973
0.9	0.932	0.973
1.0	0.932	0.932

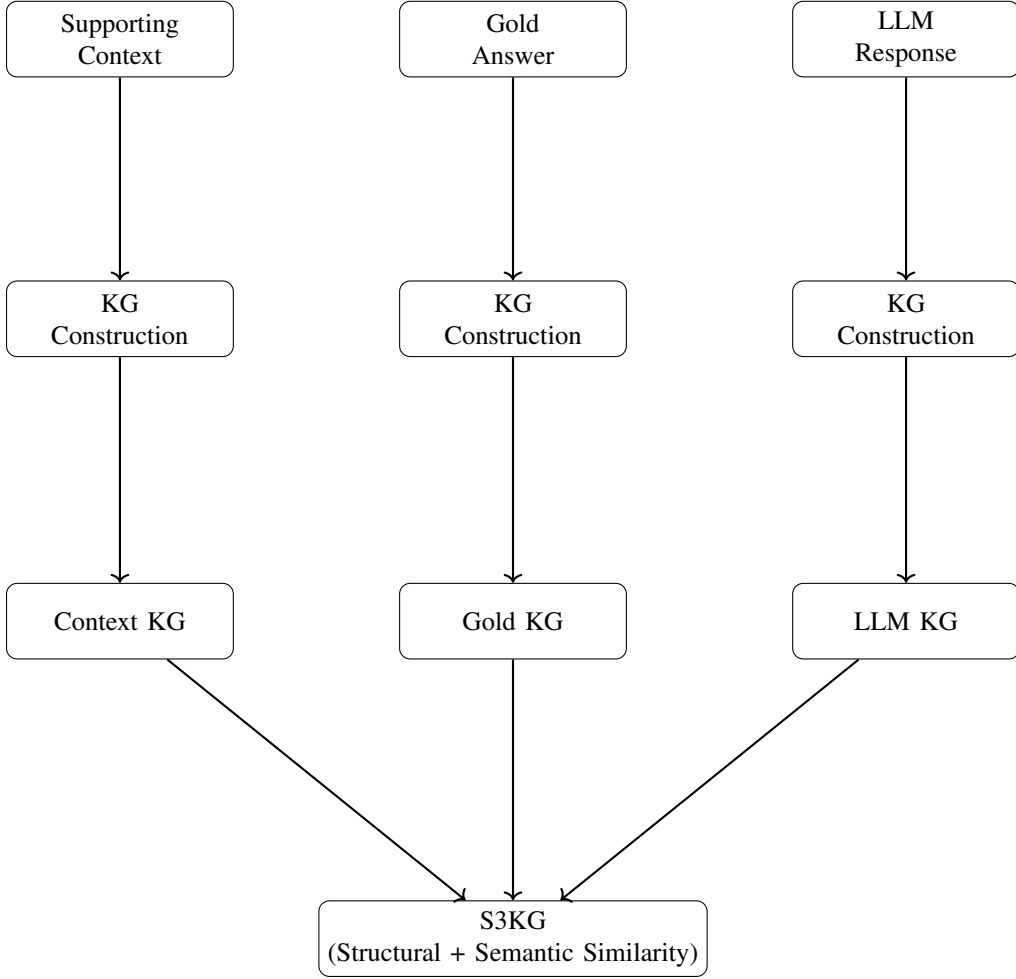


Fig. 3. Knowledge Graph construction and evaluation pipeline for contextual understanding assessment using S3KG.

TABLE XII
S3KG α SWEEP ON SK-COMBINED (F1 / AUROC).

α	F1	AUROC
0.0	0.776	0.801
0.1	0.791	0.825
0.2	0.819	0.829
0.3	0.828	0.830
0.4	0.833	0.829
0.5	0.834	0.829
0.6	0.832	0.828
0.7	0.833	0.828
0.8	0.832	0.828
0.9	0.828	0.827
1.0	0.828	0.799

TABLE XIII
S3KG α SWEEP ON SK-FINDKG (F1 / AUROC).

α	F1	AUROC
0.0	0.767	0.796
0.1	0.760	0.798
0.2	0.762	0.793
0.3	0.748	0.788
0.4	0.744	0.783
0.5	0.741	0.779
0.6	0.736	0.776
0.7	0.734	0.772
0.8	0.731	0.769
0.9	0.729	0.765
1.0	0.728	0.756

A. Short-Text Datasets

B. KG-Perturbed Paragraph Datasets

C. Wikipedia Entity-Swap (Anti-Circularity Control)

D. Knowledge Graph Construction Diagram

E. Effect of Temperature on Contextual Understanding

Tables XVII and XVIII report mean GoldSim and ContextSim respectively for each model across temperature set-

tings $T \in \{0.0, 0.3, 0.7, 1.0\}$ on both datasets. Most models show little sensitivity to temperature, with score variations within ± 0.01 – 0.02 across all settings. The exception is Falcon-7B on MesaQA, where GoldSim drops substantially from 0.6036 at $T=0.0$ to 0.4662 at $T=1.0$, indicating that higher sampling randomness significantly degrades factual alignment for this model. Scores tend to peak mildly at $T=0.3$ for

TABLE XIV
S3KG α SWEEP ON SK-GLOBI (F1 / AUROC).

α	F1	AUROC
0.0	0.811	0.898
0.1	0.883	0.934
0.2	0.888	0.937
0.3	0.889	0.936
0.4	0.891	0.936
0.5	0.890	0.935
0.6	0.892	0.935
0.7	0.892	0.935
0.8	0.892	0.935
0.9	0.892	0.934
1.0	0.892	0.917

TABLE XV
S3KG α SWEEP ON SK-OREGANO (F1 / AUROC).

α	F1	AUROC
0.0	0.777	0.884
0.1	0.803	0.892
0.2	0.811	0.892
0.3	0.812	0.892
0.4	0.812	0.892
0.5	0.810	0.892
0.6	0.812	0.892
0.7	0.812	0.891
0.8	0.810	0.891
0.9	0.810	0.891
1.0	0.810	0.770

most models before declining at $T=1.0$, suggesting that a small degree of randomness can marginally improve contextual grounding without sacrificing factual accuracy. PubMedQA scores are notably more stable across temperatures than MesaQA, likely due to the constrained nature of biomedical answers.

TABLE XVII
MEAN GOLDSIM PER MODEL ACROSS TEMPERATURES.

Dataset	Model	T=0.0	T=0.3	T=0.7	T=1.0
MesaQA	Llama-2-7B	0.6570	0.6588	0.6580	0.6406
	Gemma-7B	0.6889	0.7060	0.7015	0.6931
	Mistral-7B	0.6491	0.6567	0.6523	0.6444
	Falcon-7B	0.6036	0.6085	0.5558	0.4662
PubMedQA	Llama-2-7B	0.5220	0.5185	0.5143	0.5087
	Gemma-7B	0.5235	0.5149	0.5153	0.5204
	Mistral-7B	0.5138	0.5121	0.5085	0.5023
	Falcon-7B	0.4541	0.4330	0.4184	0.3852

TABLE XVIII
MEAN CONTEXTSIM PER MODEL ACROSS TEMPERATURES. BOLD MARKS THE BEST TEMPERATURE PER MODEL.

Dataset	Model	T=0.0	T=0.3	T=0.7	T=1.0
MesaQA	Llama-2-7B	0.7236	0.7212	0.7248	0.7026
	Gemma-7B	0.7031	0.7217	0.7045	0.7013
	Mistral-7B	0.7356	0.7585	0.7287	0.7223
	Falcon-7B	0.6458	0.6571	0.6122	0.5195
PubMedQA	Llama-2-7B	0.6567	0.6563	0.6594	0.6468
	Gemma-7B	0.6401	0.6540	0.6434	0.6319
	Mistral-7B	0.7331	0.7352	0.7314	0.7003
	Falcon-7B	0.5560	0.5165	0.5120	0.4571

TABLE XVI
S3KG α SWEEP ON WIKIPEDIA ENTITY-SWAP (F1 / AUROC / PRECISION / RECALL).

α	F1	AUROC	Precision	Recall
0.0	0.821	0.892	0.698	0.995
0.1	0.872	0.890	0.780	0.990
0.2	0.869	0.890	0.783	0.975
0.3	0.868	0.890	0.773	0.990
0.4	0.865	0.890	0.777	0.975
0.5	0.868	0.890	0.773	0.990
0.6	0.865	0.889	0.777	0.975
0.7	0.865	0.889	0.777	0.975
0.8	0.865	0.889	0.777	0.975
0.9	0.865	0.889	0.777	0.975
1.0	0.865	0.852	0.777	0.975